

MINOR PROJECT – 5

AWS Secure Infrastructure Setup IAM Access Control, VPC Networking, Security Groups and EC2 Deployment

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Project	AWS Secure Infrastructure Setup
Tools Used	AWS IAM, VPC, Subnets, Route Tables, Internet Gateway, Network ACLs, Security Groups
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This report presents the design and implementation of a secure AWS infrastructure. It covers Identity and Access Management (IAM) configuration for least-privilege access, a custom Virtual Private Cloud (VPC) with public networking components, layered network security using Security Groups and Network ACLs, and deployment verification of a web application hosted on an EC2 instance.

1. Introduction

Securing cloud infrastructure requires controlling both who can manage resources and how network traffic is allowed to flow. In this project, AWS Identity and Access Management (IAM) is used to create separate users and groups with restricted, least-privilege permissions, while a dedicated VPC isolates the project's networking resources. A public subnet, internet gateway and route table provide external connectivity, and two independent security layers — Security Groups (instance-level, stateful) and Network ACLs (subnet-level, stateless) — restrict inbound and outbound traffic. Finally, an EC2 instance was launched inside the secured subnet and a sample web page was deployed and verified over its public IP address.

2. Objectives

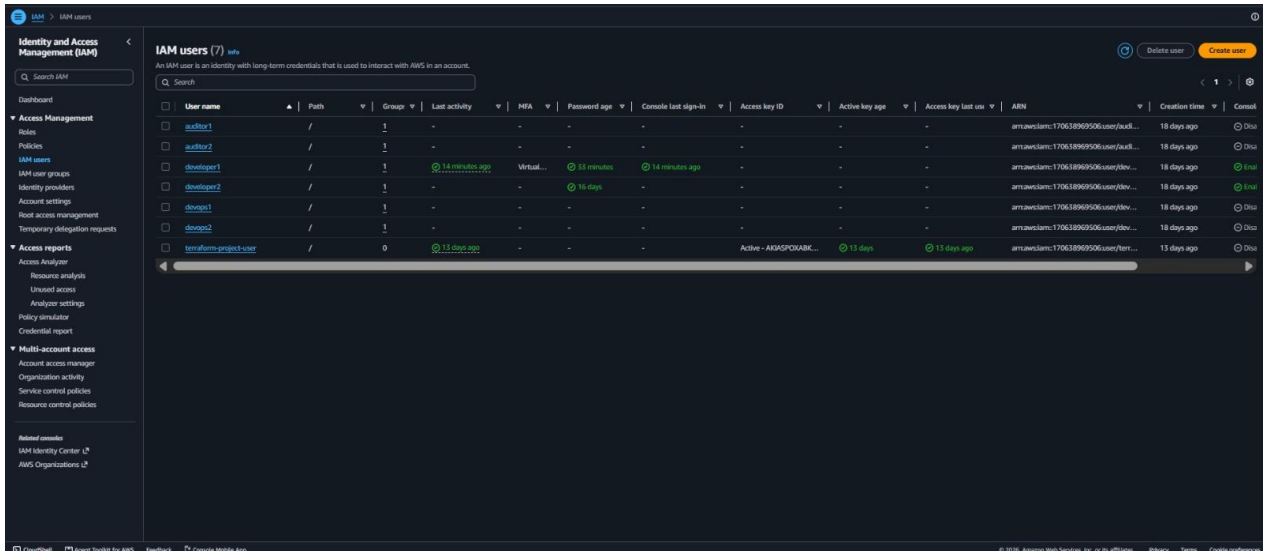
- Create dedicated IAM users and organise them into functional groups (auditors, developers, devops).
- Apply a custom least-privilege IAM policy restricting EC2 termination and IAM modification.
- Design a custom VPC with a public subnet, route table and internet gateway.
- Configure a Network ACL to allow only SSH, HTTP and HTTPS traffic to the subnet.
- Configure a Security Group to allow SSH access only from a trusted IP and open HTTP to the public.
- Launch an EC2 instance inside the secured public subnet.
- Deploy and verify a sample web page over the instance's public IP address.

3. Project Configuration

Component	Configuration
VPC	Project-VPC (10.0.0.0/16)
Subnet	Public-Subnet (10.0.1.0/24), Availability Zone ap-south-1a
Internet Gateway	Project-IGW (attached to Project-VPC)
Route Table	Public-Route-Table — 0.0.0.0/0 → Internet Gateway, 10.0.0.0/16 → local
Network ACL	Allows SSH (22) from trusted IP, HTTP (80) and HTTPS (443) from anywhere; denies all
Security Group	Project-EC2-SG — SSH (22) restricted to trusted IP, HTTP (80) open, all outbound
EC2 Instance	Project-EC2 (t3.micro), Public IP 43.205.112.214, Subnet: Public-Subnet
IAM Groups	Developers (developer1, developer2) with Developers-S3-Policy and Project-Developer-E
IAM Users	auditor1, auditor2, developer1, developer2, devops1, devops2, terraform-project-user

4. IAM Implementation Evidence

Seven IAM users were created to represent distinct roles within the project — auditors, developers and DevOps engineers — plus a dedicated user for Terraform automation. Each user is scoped to only the access required for their role, following the principle of least privilege.

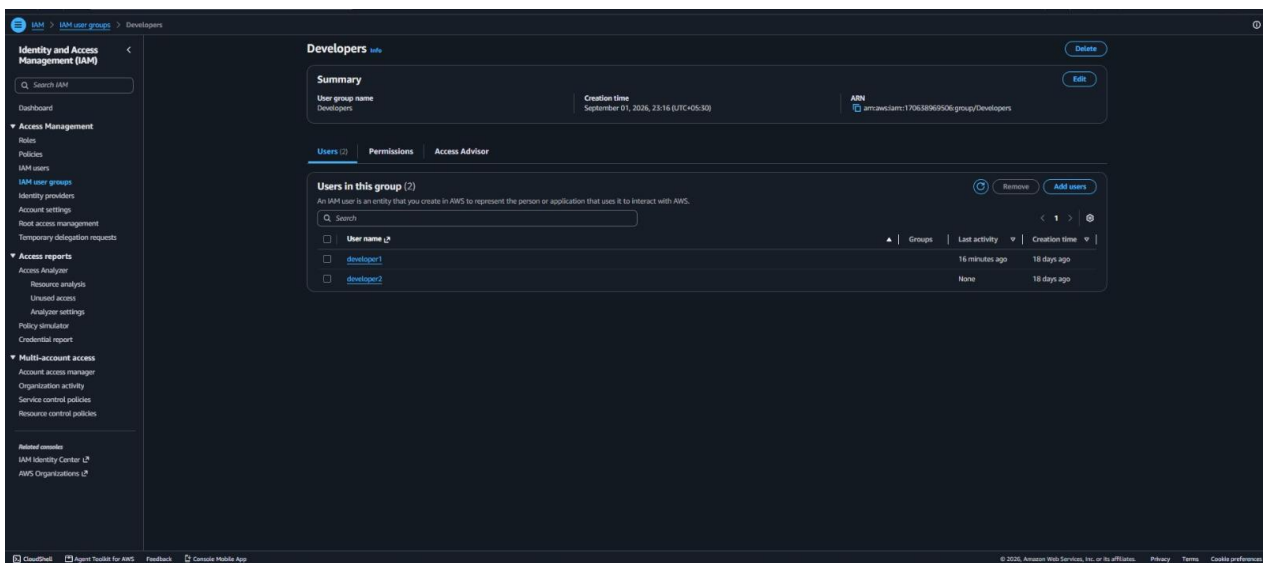


The screenshot shows the AWS IAM console 'IAM users (7)' page. It displays a table of IAM users with columns for User name, Path, Group, Last activity, MFA, Password age, Console last sign-in, Access key ID, Active key age, Access key last use, ARN, Creation time, and Console. The users listed are auditor1, auditor2, developer1, developer2, devops1, devops2, and terraform-project-user. developer1 and developer2 are grouped under the 'Developers' group.

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Access key last use	ARN	Creation time	Console
auditor1	/	1	-	-	-	-	-	-	-	arn:aws:iam::170638969506:user/aud1...	18 days ago	Dis
auditor2	/	1	-	-	-	-	-	-	-	arn:aws:iam::170638969506:user/aud2...	18 days ago	Dis
developer1	/	1	14 minutes ago	Virtual...	35 minutes	14 minutes ago	-	-	-	arn:aws:iam::170638969506:user/dev...	18 days ago	Enr
developer2	/	1	-	-	16 days	-	-	-	-	arn:aws:iam::170638969506:user/dev...	18 days ago	Enr
devops1	/	1	-	-	-	-	-	-	-	arn:aws:iam::170638969506:user/dev...	18 days ago	Dis
devops2	/	1	-	-	-	-	-	-	-	arn:aws:iam::170638969506:user/dev...	18 days ago	Dis
terraform-project-user	/	0	13 days ago	-	-	-	Active - AKASPOXAM...	13 days	13 days ago	arn:aws:iam::170638969506:user/ter...	13 days ago	Dis

Figure 1: IAM users — seven role-based users created for the project.

The developer1 and developer2 users were placed in a Developers group. Grouping users allows permissions to be managed centrally rather than attached to each user individually.



The screenshot shows the AWS IAM console 'IAM user groups' page for the 'Developers' group. It displays a summary of the group, including its name, creation time, and ARN. Below the summary, there is a section for 'Users in this group (2)' which lists the users developer1 and developer2.

User name	Groups	Last activity	Creation time
developer1		16 minutes ago	18 days ago
developer2		None	18 days ago

Figure 2: Developers group containing developer1 and developer2.

The Developers group has two customer-managed policies attached: Developers-S3-Policy for S3 access, and Project-Developer-EC2-Policy, a custom policy limiting EC2 permissions.

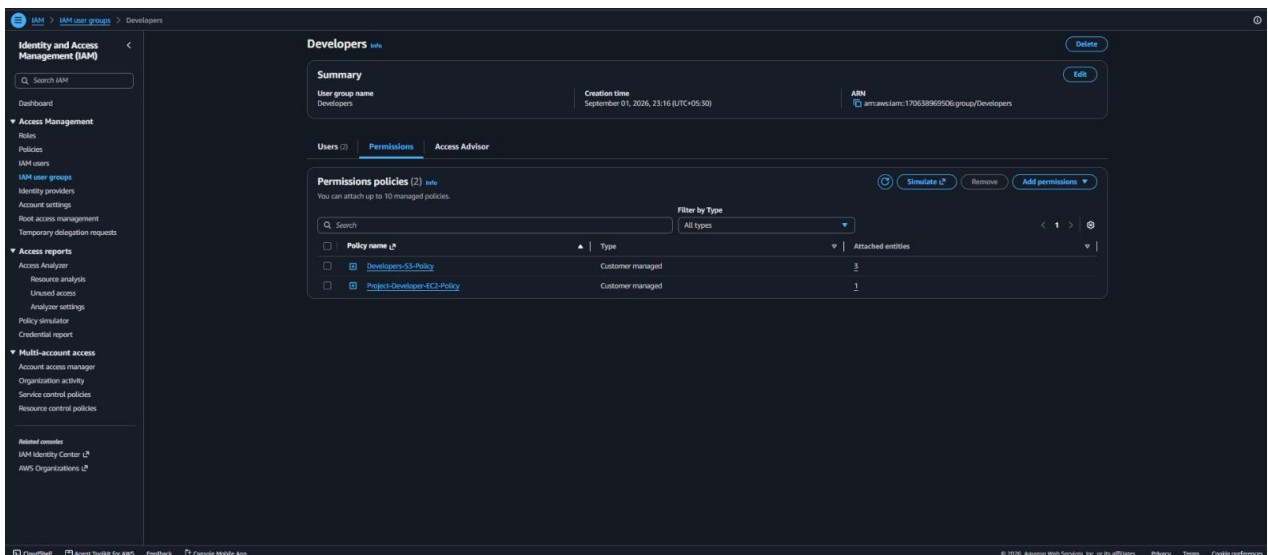


Figure 3: Permissions attached to the Developers group.

The Project-Developer-EC2-Policy explicitly denies EC2 write actions such as termination and all IAM write actions, while allowing limited list and write access needed for day-to-day EC2 operations. This prevents developers from deleting instances, VPCs, security groups, or IAM users, even though they retain the access needed to do their work.

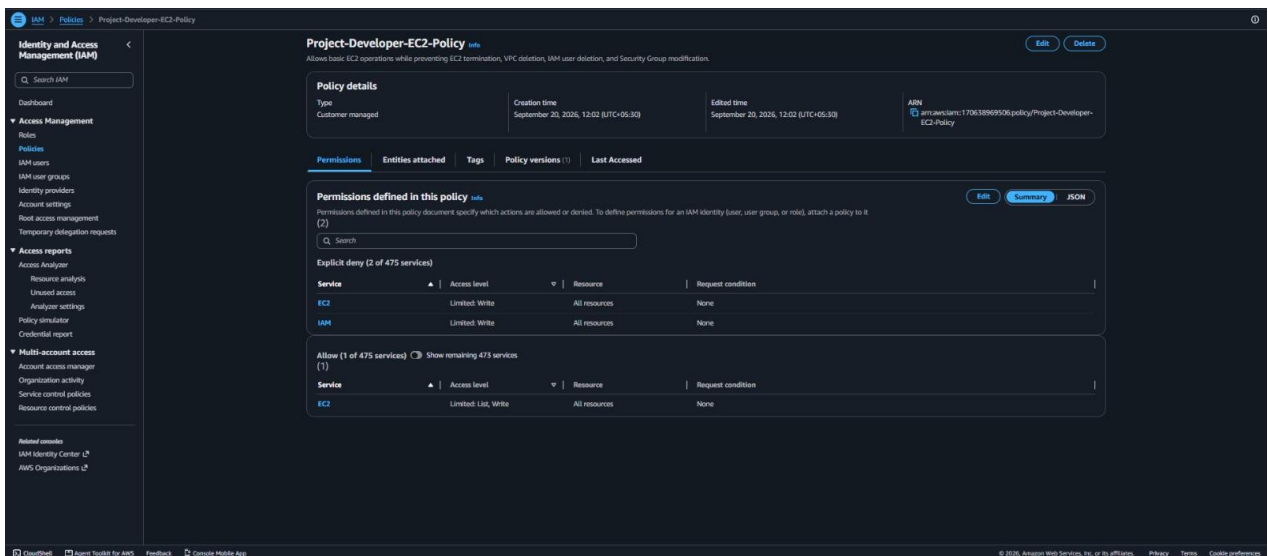


Figure 4: Project-Developer-EC2-Policy — explicit deny on EC2 termination and IAM changes.

5. VPC and Networking Implementation Evidence

A dedicated VPC named Project-VPC was created with the CIDR block 10.0.0.0/16, separate from the account's default VPC and the Terraform-managed VPC used in an earlier project.

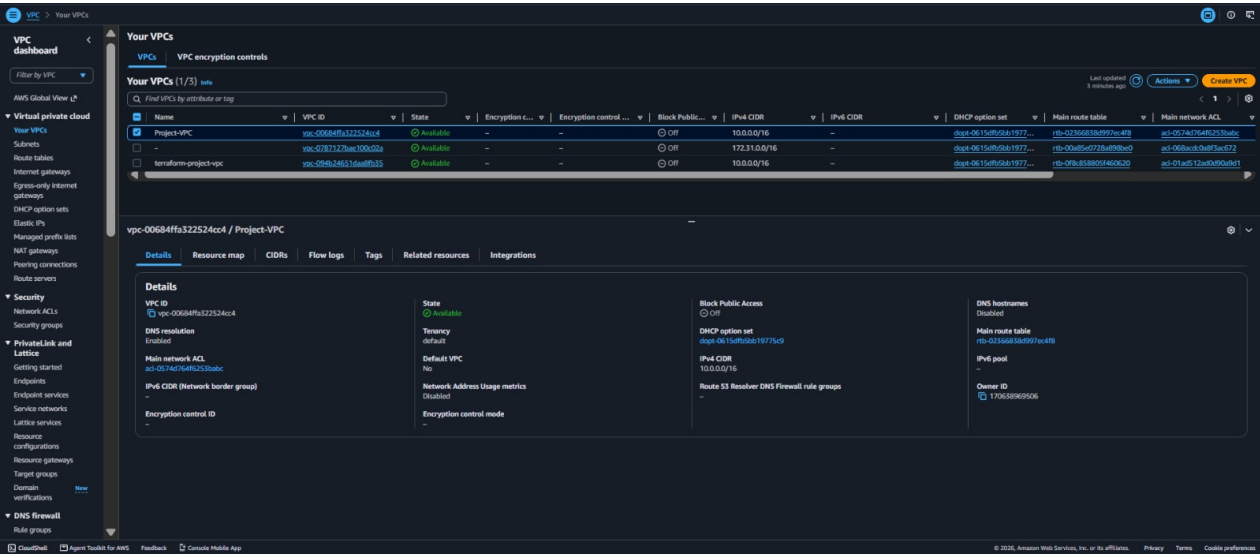


Figure 5: Project-VPC (10.0.0.0/16) details.

A public subnet (10.0.1.0/24) was created inside Project-VPC in the ap-south-1a availability zone, configured to auto-assign public IPv4 addresses so that instances launched in it are reachable from the internet.

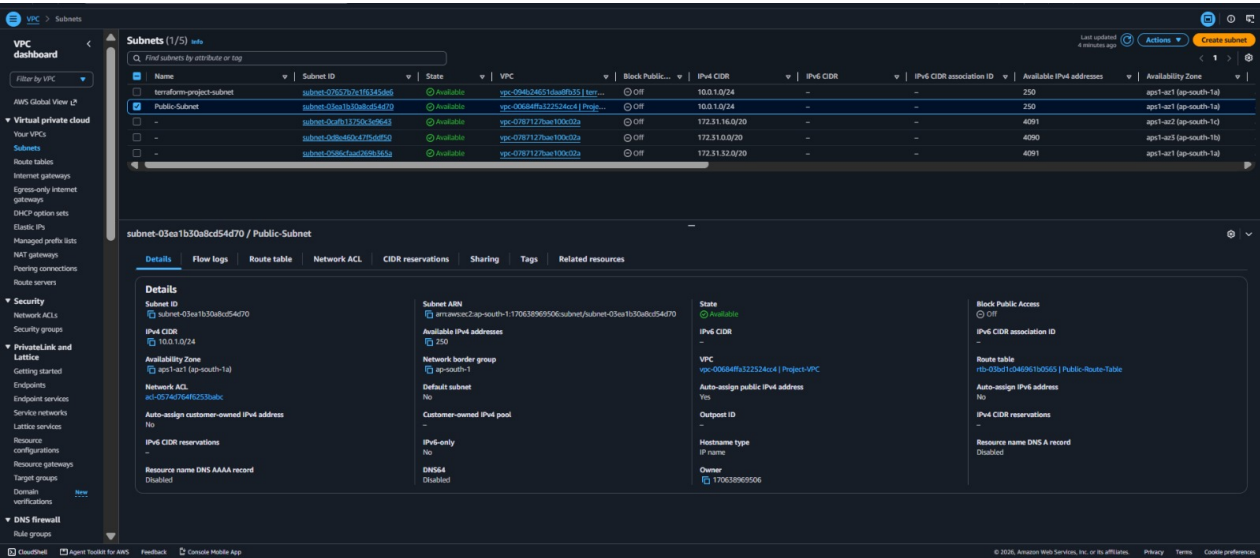


Figure 6: Public-Subnet (10.0.1.0/24) with auto-assign public IPv4 enabled.

An internet gateway, Project-IGW, was created and attached to Project-VPC to allow traffic between the VPC and the public internet.

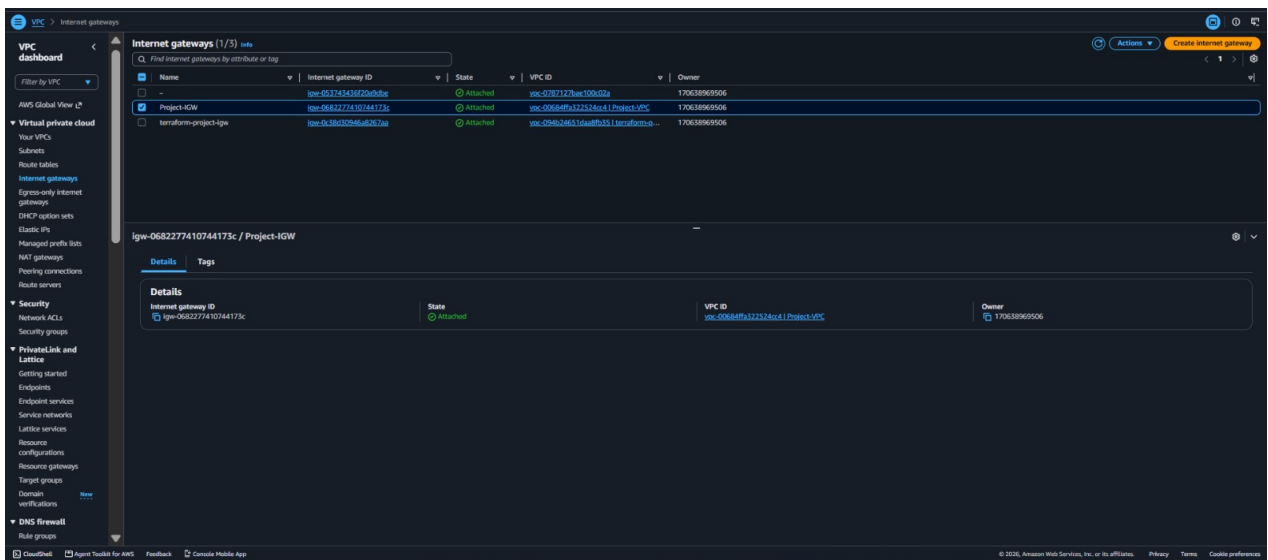


Figure 7: Project-IGW attached to Project-VPC.

A custom route table, Public-Route-Table, was associated with the public subnet. It routes all external traffic (0.0.0.0/0) to the internet gateway and keeps traffic within the VPC's own CIDR block (10.0.0.0/16) local.

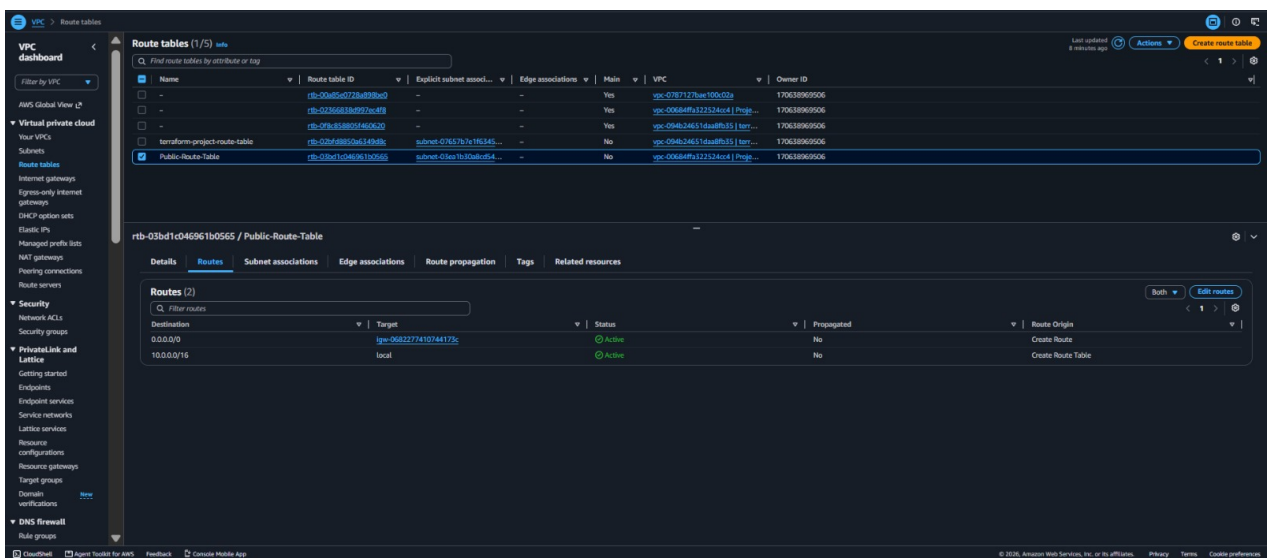


Figure 8: Public-Route-Table with a default route to the internet gateway.

6. Network ACL Configuration Evidence

The Network ACL associated with the public subnet was configured with explicit inbound rules permitting SSH (port 22) only from a single trusted IP address, plus HTTP (port 80) and HTTPS (port 443) from any source. All other inbound traffic is denied by the default rule.

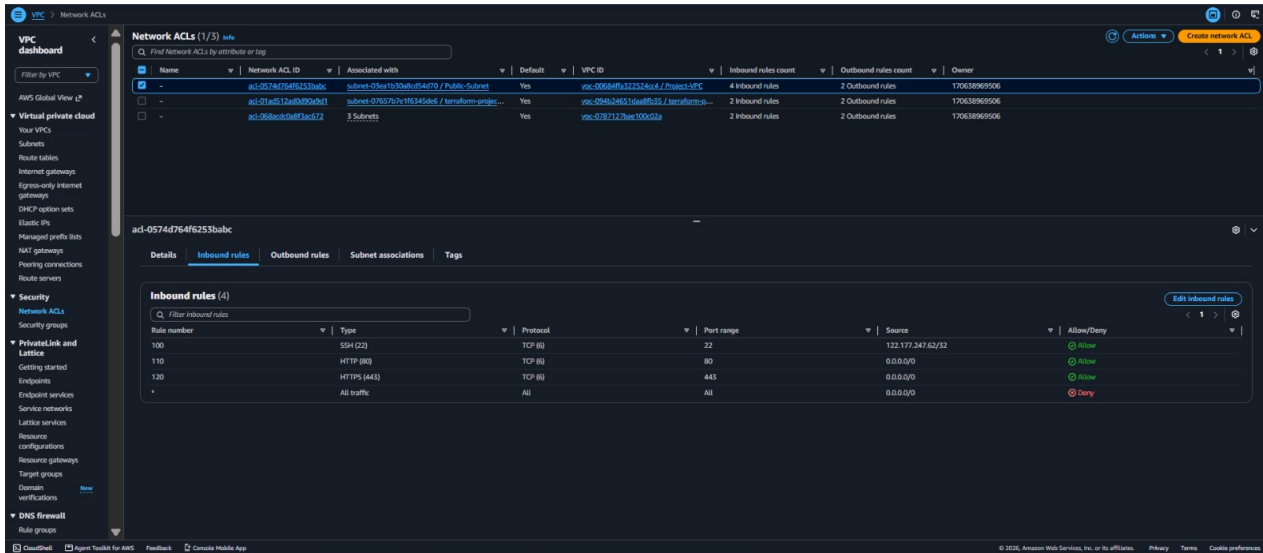


Figure 9: Network ACL inbound rules — SSH restricted, HTTP/HTTPS open, all else denied.

Outbound rules on the same Network ACL allow all outbound traffic, which is required so that response traffic for the permitted inbound connections can leave the subnet, since Network ACLs are stateless and evaluate each direction independently.

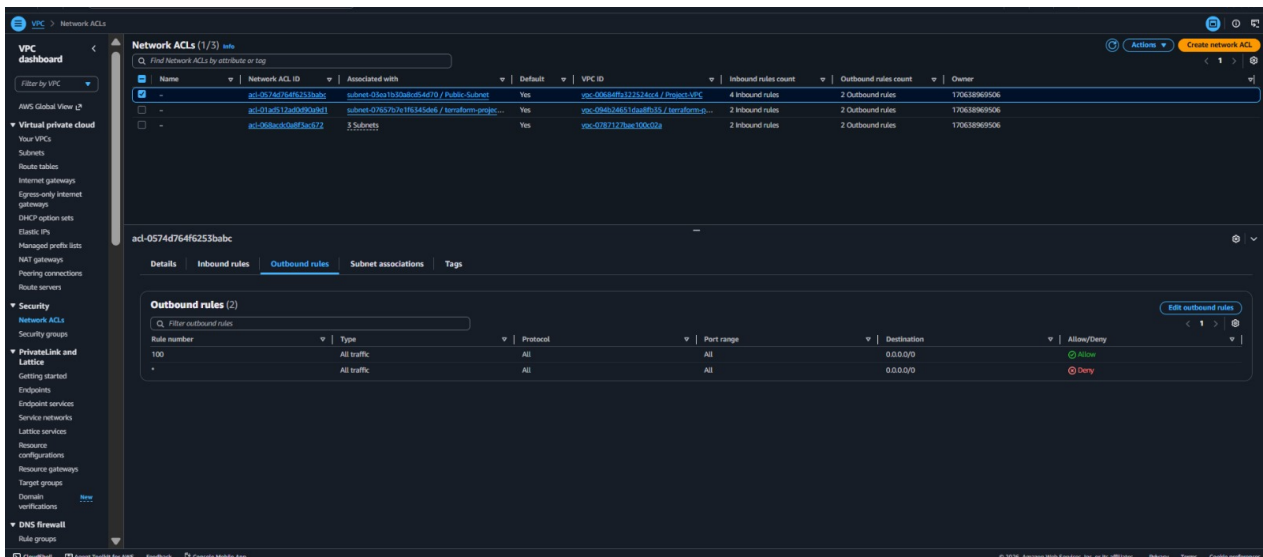


Figure 10: Network ACL outbound rules — all outbound traffic allowed.

7. Security Group Configuration Evidence

Project-EC2-SG is the security group attached to the EC2 instance. Its inbound rules allow SSH (port 22) only from the same trusted IP address used in the Network ACL, and HTTP (port 80) from any source so the deployed web page is publicly reachable.

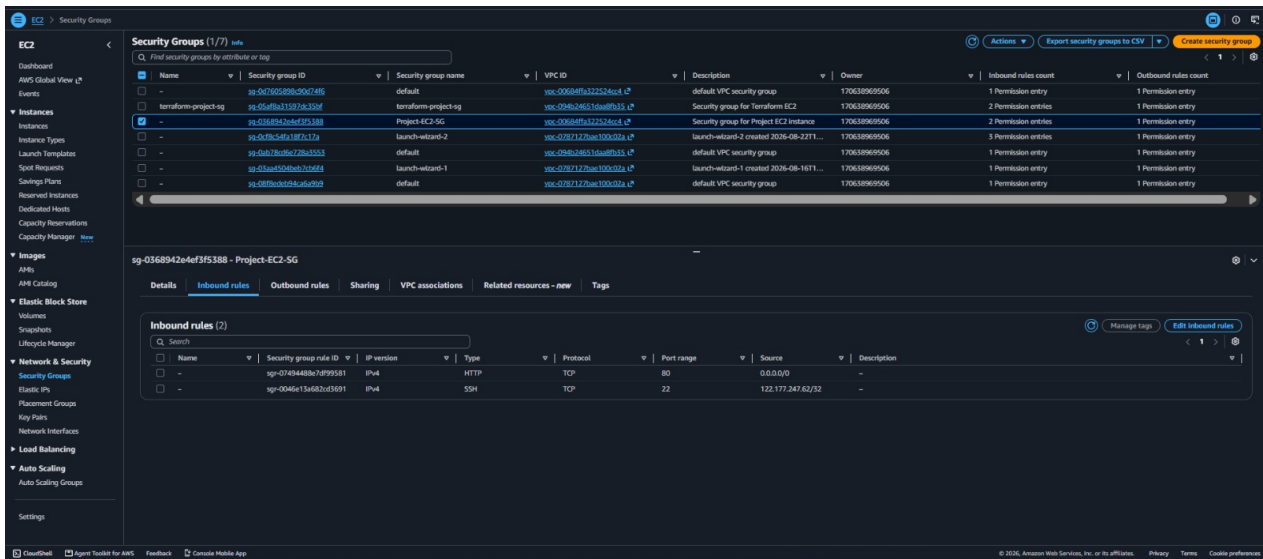


Figure 11: Project-EC2-SG inbound rules — SSH restricted, HTTP open to the public.

Because security groups are stateful, a single outbound rule permitting all outbound traffic is sufficient to allow return traffic for any permitted inbound connection.

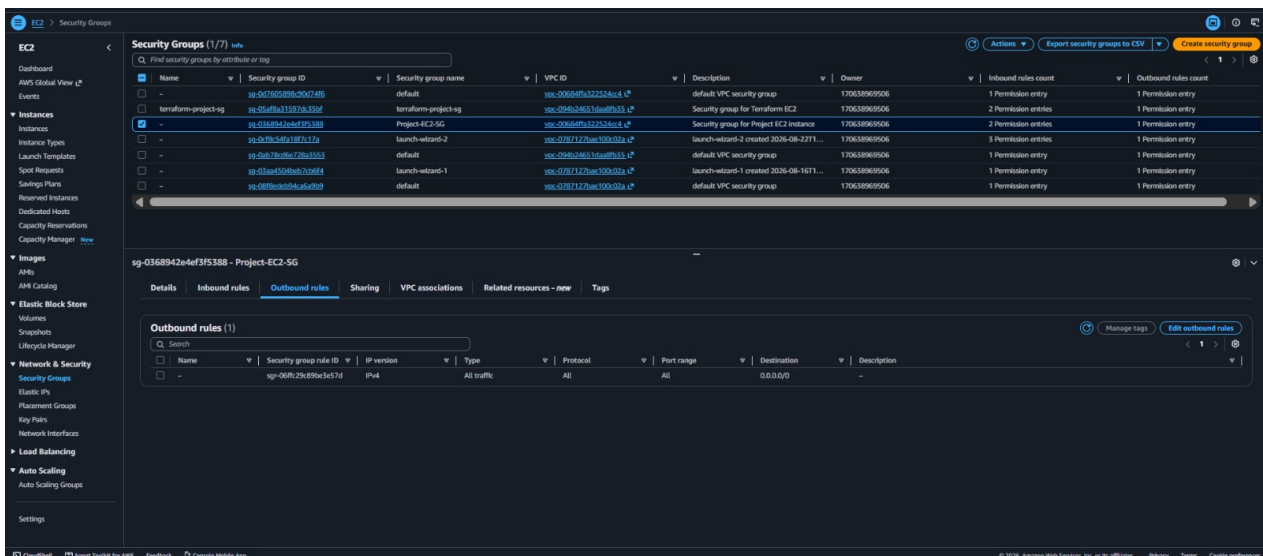


Figure 12: Project-EC2-SG outbound rules — all outbound traffic allowed.

8. EC2 Instance and Deployment Verification

The Project-EC2 instance (t3.micro) was launched inside the Public-Subnet of Project-VPC, using the Project-EC2-SG security group. The instance is in the Running state with the public IPv4 address 43.205.112.214.

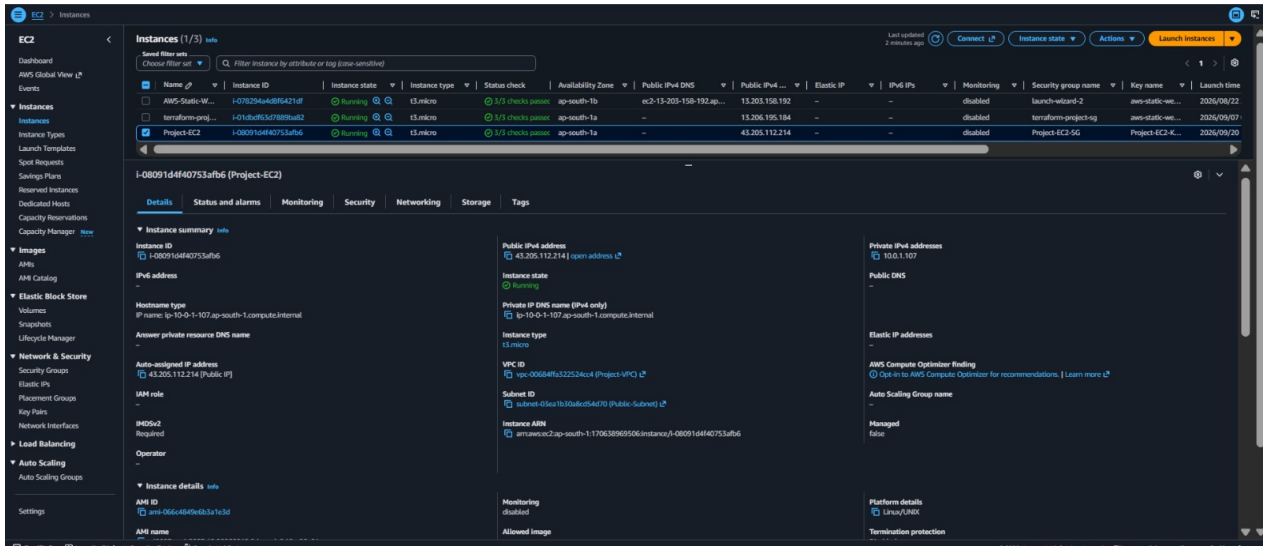


Figure 13: Project-EC2 instance running in Public-Subnet with public IP 43.205.112.214.

The deployment was verified by browsing to the instance's public IP address, which served the expected sample page, confirming that the VPC, subnet, routing, internet gateway, Network ACL and Security Group were all correctly configured to allow HTTP access to the instance.



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Figure 14: Web page served successfully from the EC2 instance's public IP address.

9. Validation Summary

The infrastructure was validated end-to-end, from IAM access boundaries through to a publicly reachable web page:

- IAM users created and organised into a Developers group with least-privilege policies.
- Custom IAM policy verified to deny EC2 termination and IAM modification for developers.
- Custom VPC (Project-VPC, 10.0.0.0/16) created with a public subnet (10.0.1.0/24).
- Internet gateway attached and route table configured with a default route to it.
- Network ACL configured to allow only SSH (trusted IP), HTTP and HTTPS inbound.
- Security Group configured to allow SSH (trusted IP) and HTTP, with all outbound allowed.
- EC2 instance launched successfully in the public subnet with a public IP address.
- Sample web page verified as reachable over HTTP from the instance's public IP.

10. Conclusion

This project successfully demonstrates a secure, layered AWS infrastructure combining least-privilege IAM access control with a purpose-built VPC network protected by both Network ACLs and Security Groups. Role-based IAM users and a restrictive custom policy limit what developers can change, while the network layer permits only the traffic required for administration (SSH from a trusted IP) and public web access (HTTP). The successful launch of the EC2 instance and verified delivery of the sample web page confirm that every layer of the configuration — IAM, routing, Network ACLs and Security Groups — was implemented correctly and works together as intended.

11. References

- AWS Identity and Access Management (IAM) documentation
- Amazon VPC documentation
- Amazon EC2 documentation
- AWS Security Groups and Network ACLs documentation

— End of Report —